

Homework 1

Selected Topics in Numerical Mathematics 2025–2026

1. Generalization of the Remez algorithm

The exercise is about a generalization of the Remez algorithm from the space of polynomials of degree $\leq n$ to the space

$$S := \mathcal{Lin}\{1, x^2, x^4\}.$$

Show first that the system of functions $\{1, x^2, x^4\}$ forms a Chebyshev system of functions on interval $[a, b]$, $a > 0$. Then write in Matlab the function

```
function sol = RemezSpecial(fun,a,b,tol,kmax),
```

which via the Remez algorithm with the tolerance `tol` finds an element of best uniform approximation for the function `fun` on the interval $[a, b]$ from the approximation space S . Here `kmax` denotes the maximal number of steps of the Remez algorithm and `tol` is the prescribed tolerance. Function should return a vector of coefficients, which defines the best approximant $\tilde{f}(x) \in S$.

Test the function on the following data:

Find the best uniform approximant $\tilde{f}$ for the function

$$f(x) = \exp\left(\frac{3}{5}x\right) \cos\left(\frac{2}{5}\pi x\right), \quad (1)$$

on the interval $[1, 3]$ with the tolerance `tol` = 10^{-3} and `kmax`=10.

Advice: Use functions `Remez.m` and `RemezStep.m`, which you implemented at tutorials.

2. Least squares method in the space of polynomials

Let $\mathbf{x} = (x_i)_{i=1}^N$ be the vector of points, such that $a \leq x_1 < x_2 < \dots < x_N \leq b$. On $L^2([a, b])$ we define the following inner product

$$\langle f, g \rangle := \sum_{i=1}^N f(x_i)g(x_i). \quad (2)$$

Our goal is to find a polynomial of degree $\leq n$, which best approximates given function $f \in L^2([a, b])$ with respect to the inner product (2).

In Matlab write the function

`function sol = leastSquares(f, x, n),`

which returns the coefficients of the best least squares polynomial approximant with respect to the inner product (2) and points $\mathbf{x}$.

Test the function on the following data:

We approximate function (1). Take

$$[a, b] = [1, 4] \quad \text{and} \quad \mathbf{x} = \left(1, \frac{6}{5}, \frac{7}{5}, \dots, \frac{19}{5}, 4\right).$$

Find the best least squares polynomial approximants of degree 4 and 5 for the inner product (2).

3. Orthogonal polynomials

Let $\mathbf{x} = (x_i)_{i=1}^N$ be the vector of points, such that $a \leq x_1 < x_2 < \dots < x_N \leq b$. In Matlab write the function

`function P = orthogonal(x, n) ,`

which returns a matrix P with rows representing the coefficients of the first n orthogonal polynomials with respect to the inner product (2) and points $\mathbf{x}$.

Test the function on the following data:

Let

$$[a, b] = [0, 2\pi] \quad \text{and} \quad \mathbf{x} = \left(0, \frac{\pi}{4}, \frac{\pi}{2}, \dots, \frac{7\pi}{4}, 2\pi\right).$$

Find the first seven orthogonal polynomials.